Long Range (Open Plains / Maximum-Distance Combat)

Executive Summary

Long Range is a **wide-open plains battlefield** built for **maximum-distance engagements, extreme sightlines, long-range artillery duels, drone reconnaissance, and autonomous maneuvering across unobstructed terrain**. The environment features rolling grasslands, sparse tree lines, shallow ridges, abandoned farm structures, wind farms, dry creek beds, and isolated observation towers. The map emphasizes Obsidian Protocol's autonomous long-range targeting logic, sensor-fusion under heat shimmer and dust, comms-relay warfare, and logistics across vast, open terrain. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of open-plains terrain. Key features include extreme sightlines, long-range fire corridors, drone-heavy recon, and maneuver warfare across minimal cover

Design Goals & Gameplay

Maximum-Distance Combat

Long Range prioritizes engagements at the absolute limits of weapon range. Artillery, missile vehicles, snipers, and long-range drones dominate.

Open-Terrain Maneuver Warfare

Minimal cover forces reliance on speed, spacing, and autonomous movement. Units must avoid predictable lines of advance.

Recon & Counter-Recon

Drones, sensor towers, and long-range optics define the information war. Destroying recon assets creates massive blind zones.

Autonomy & Distance-Aware Pathfinding

Autonomous units choose routes based on elevation, cover scarcity, heat shimmer, and sensor clarity. Units avoid exposed flats during peak heat unless necessary.

Sensor & Information Warfare

Heat distortion, dust haze, mirage effects, and long-range interference degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates blind pockets across the plains.

Logistics & Resources

Fuel depots, farm structures, and wind-farm maintenance yards serve as resupply nodes. Convoys must navigate exposed roads and avoid long-range ambush corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between artillery dominance, recon superiority, or fast armor. Blue deploys from the southwest dry-creek staging zone; Red deploys from the northeast ridge-top command.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² open-plains zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Dry-Creek Staging Zone) — (0,0), 150×150 m Cargo crates, armored vehicles, drone pads.
Red HQ (Ridge-Top Command) — (1000,1000), 150×150 m Elevated bunker with radar mast, AA emplacements, and long-range sensors.
Central Plains Basin — (500,450), 300×300 m Wide open flats, extreme sightlines, heat shimmer, artillery duels.
Wind Farm Depot — (250,800), 120×120 m Turbines, maintenance sheds, interior flanking routes.
Ridge Network — (800,300), 200×200 m High-ground overwatch, sniper nests, drone launch pads.
Long-Range Fire Corridor — (0,450 → 400,900) Primary engagement zone; extreme range, minimal cover.
Service Road Route — (300,0 → 450,350) Safer but predictable; vulnerable to overwatch from ridges.
Ridge Relay A — (650,150), +240 m Long-range sensor vantage point.
Ridge Relay B — (150,650), +260 m Comms relay hub and recon-control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Long-Range Fire Corridor

Major engagement zone; extreme sightlines and maximum-range duels.

Ridge Network

High-ground overwatch; ideal for snipers, artillery, and drones.

Plains Basin

Long-range engagements distorted by heat shimmer and dust haze.

Service Road Route

Predictable but safe; vulnerable to ridge overwatch.

Sightlines

Ridge Relay A provides long-range sensor coverage (~400 m). Heat distortion and dust haze create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW dry-creek staging + armor.
Red Deployment: NE ridge-top command + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Plains, ridges, dry creek beds, wind farms, farm structures.

Vegetation

Sparse grass, shrubs, scattered trees.

Buildings & Props

Military: AA guns, radar dishes, bunkers. Industrial: Turbines, sheds, pipelines. Logistics: Trucks, crates, fuel drums, drones.

Effects

Dust haze, heat shimmer, mirage distortion, wind gusts.

Audio

Wind, distant machinery, radio chatter, drone hum.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Plains/Rid	3	200k	600k
Industrial/	4	80k	320k
Buildings/	15	50k	750k
Roads/Pat	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out plains basin, ridges, wind farm, AR anchor test.
2. **Core Systems — Month 2** NavMesh, long-range autonomy behaviors, recon logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (dust haze, heat shimmer).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify long-range targeting, recon-aware pathfinding, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under heat distortion.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, long-range logic, deployment budget enforcement, anchor drift, sensor accuracy.